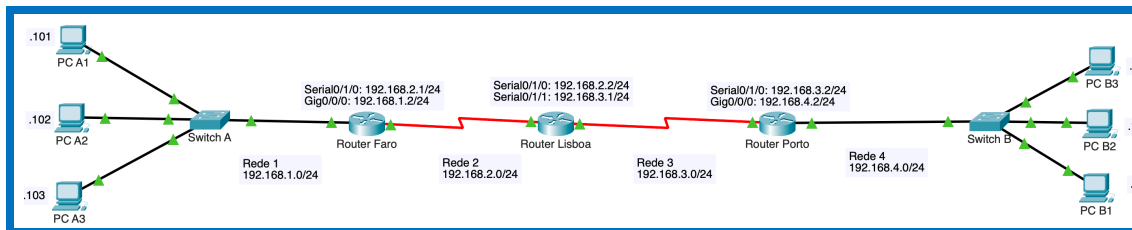


Context

This activity aims to test the use of a link state protocol (OSPF) for the automatic construction of forwarding tables for various routers. Only one OSPF area will be considered in this activity.

1 – Basic topology to be implemented in Packet Tracer

Area 0



The proposed architecture can be downloaded from the following link, where some of the basic configurations for the IP addressing of the physical interfaces have already been made.

In addition to the physical interfaces, loopback interfaces are also to be configured on each router. For this purpose, the following addresses must be assigned:

- Router Faro: 172.16.1.1
- Router Lisboa: 172.16.2.1
- Router Porto: 172.16.3.1

The use of loopback interfaces has the advantage that they never fail, unlike physical interfaces. This stability ensures that the RID value of each router will never change, since, if not manually defined, it is determined by the highest IP address of the configured loopback interfaces and, if these do not exist, by the highest IP address of the physical interfaces.

If the loopback interface is not active, if a physical interface becomes inactive, a new DR and BDR election process must take place. However, it is important to note that in P2P connections, such as the architecture proposed in this activity, it is not necessary to elect a DR/BDR because there are no alternative paths.

2 – OSPF Configuration

The OSPF configuration is performed on each router, with a view to indicating which available interfaces should participate in the OSPF process. To enter routing configuration mode, use the command `router OSPF process_number`. The process number is a local parameter for the router and, as such, may vary from router to router within the same area. In this case, OSPF process numbers 100, 200, and 300 should be used for the Faro, Lisboa, and Porto routers, respectively.

Next, indicate the interfaces you want to include in OSPF and the area to which they belong with the command `network IP_address network_mask area area_number`. All interfaces identified by the command become part of the OSPF updates.

The network mask is based on wildcards, where a 0 indicates that the corresponding IP address byte must match and 255 indicates that the IP address byte can be any. In general, as there are few interfaces, it is simpler to specify the address of all interfaces with the mask 0.0.0.0.

Faro router configuration

First, enter OSPF routing configuration mode by specifying the number 100 for the process (it can be any number between 1 and 65535). Next, specify the addresses of the router interfaces with the mask 0.0.0.0, since you want OSPF to identify exactly those interfaces. Each interface involved in the OSPF process will advertise its respective network. Additionally, loopback interfaces are also included in the OSPF process.

```
Router> en
Router# conf t
Router-Faro (config)# int loopback 0
Router-Faro (config-if)# ip address 172.16.1.1 255.255.255.0
Router-Faro (config-if)# no shut down
Router-Faro (config-if)# ^Z
Router (config)# router ospf 100
Router-Faro (config-router)# network 192.168.2.1 0.0.0.0 area 0
Router-Faro (config-router)# network 192.168.1.2 0.0.0.0 area 0
Router-Faro (config-router)# network 172.16.1.1 0.0.0.0 area 0
Router-Faro (config-router)# ^Z
```

Lisboa router configuration

In this case, the interfaces covered by OSPF include 192.168.2.1 and 192.168.3.2. As an example, a mask selection is used that allows both interfaces to be configured with a single command.

```
Router> en
Router# conf t
Router- Lisboa (config)# int loopback 0
Router- Lisboa (config-if)# ip address 172.16.2.1 255.255.255.0
```

```

Router- Lisboa (config-if)# no shut down
Router- Lisboa (config-if)# ^Z
Router (config)# router ospf 200
Router-Lisboa (config-router)# network 192.168.0.0 0.0.255.255 area 0
Router-Faro (config-router)# network 172.16.2.1 0.0.0.0 area 0
Router- Lisboa (config-router)# ^Z

```

Check if a message like this is returned: 01:03:28: %OSPF-5-ADJCHG: Process 200, Nbr 172.16.1.1 on Serial0/1/0 from LOADING to FULL, Loading Done, noting that the process of establishing a neighbor relationship and subsequent adjacency has already converged to the final state "FULL."

Porto router configuration

The configuration of the Porto router, which is symmetrical to that of the Faro router, follows a similar configuration, but for the respective networks. As an example, in this case, we have chosen to configure the network mask differently, but in a way that fulfills the same objectives.

```

Router> en
Router# conf t
Router-Porto (config)# int loopback 0
Router-Porto (config-if)# ip address 172.16.3.1 255.255.255.0
Router-Porto (config-if)# no shut down
Router-Porto (config-if)# ^Z
Router-Porto (config)# router ospf 300
Router-Porto (config-router)# network 192.168.3.0 0.0.0.255 area 0
Router-Porto (config-router)# network 192.168.4.0 0.0.0.255 area 0
Router-Faro (config-router)# network 172.16.3.1 0.0.0.0 area 0
Router-Porto (config-router)# ^Z

```

3 – Verification of routing tables

Using the show ip route command, check the resulting routing tables, noting the routes learned by OSPF with the code "O."

Check the respective routing tables for the metrics associated with the routes learned with the OSPF protocol, which contain two parameters:

Administrative Distance (AD): is a measure of the trustworthiness of a routing source. Lower numbers are more trusted. For OSPF, the AD is always 110. This value is standard and is used to compare OSPF routes to routes learned from other protocols (e.g., EIGRP, BGP).

OSPF Metric (Cost): is used by OSPF to determine the best path to a destination network. The cost is based on the cumulative cost of all outgoing interfaces along the path to the destination network. By default, OSPF calculates the cost based on the bandwidth of the interfaces. The formula is $\text{Cost} = \text{Reference Bandwidth} / \text{Interface Bandwidth}$. The default reference bandwidth is 100 Mbps.

```
Router-Faro# show ip route
```

```
Router-Faro#sh ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
      172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
C       172.16.1.0/24 is directly connected, Loopback0
L       172.16.1.1/32 is directly connected, Loopback0
O       172.16.2.1/32 [110/65] via 192.168.2.2, 00:03:46, Serial0/1/0
O       172.16.3.1/32 [110/129] via 192.168.2.2, 00:02:00, Serial0/1/0
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.1.2/32 is directly connected, GigabitEthernet0/0/0
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, Serial0/1/0
L       192.168.2.1/32 is directly connected, Serial0/1/0
O       192.168.3.0/24 [110/128] via 192.168.2.2, 00:33:20, Serial0/1/0
O       192.168.4.0/24 [110/129] via 192.168.2.2, 00:24:26, Serial0/1/0
```

In this case, network 3 has a metric of 128. This is because serial connections have a bandwidth of 1,544 Mbps, so $100,000,000/1,544,000$, which gives an approximate cost of 64. Therefore, the combination of networks 2 and 3 has a total cost of 128. In the case of network 4, a fast Ethernet section must be added, which has a cost of 1.

```
Router-Lisboa# show ip route
```

```
Router-Lisboa#sh ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
      172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
O       172.16.1.1/32 [110/65] via 192.168.2.1, 00:05:19, Serial0/1/0
C       172.16.2.0/24 is directly connected, Loopback0
L       172.16.2.1/32 is directly connected, Loopback0
O       172.16.3.1/32 [110/65] via 192.168.3.2, 00:02:59, Serial0/1/1
O       192.168.1.0/24 [110/65] via 192.168.2.1, 00:26:39, Serial0/1/0
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, Serial0/1/0
L       192.168.2.2/32 is directly connected, Serial0/1/0
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, Serial0/1/1
L       192.168.3.1/32 is directly connected, Serial0/1/1
O       192.168.4.0/24 [110/65] via 192.168.3.2, 00:25:25, Serial0/1/1
```

In this case, the cost of 65 comes from the combination of network 3 and network 4.

```

Router-Porto# show ip route
Router-Porto#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
O       172.16.1.1/32 [110/129] via 192.168.3.1, 00:05:48, Serial0/1/0
O       172.16.2.1/32 [110/65] via 192.168.3.1, 00:05:14, Serial0/1/0
C       172.16.3.0/24 is directly connected, Loopback0
L       172.16.3.1/32 is directly connected, Loopback0
O      192.168.1.0/24 [110/129] via 192.168.3.1, 00:27:08, Serial0/1/0
O      192.168.2.0/24 [110/128] via 192.168.3.1, 00:34:48, Serial0/1/0
        192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, Serial0/1/0
L       192.168.3.2/32 is directly connected, Serial0/1/0
        192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.4.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.4.2/32 is directly connected, GigabitEthernet0/0/0

```

In this case, the costs of 128 and 129 follow the same explanation as for the Faro router.

4 – Verifying the OSPF configuration

Using the `show ip protocols` command, you can confirm the protocols configured on each router, in particular to confirm the desired parameters in the OSPF configuration.

```
Router-Faro# show ip protocols
```

```
Routing Protocol is "ospf 100"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 172.16.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.2.1 0.0.0.0 area 0
    192.168.1.2 0.0.0.0 area 0
    172.16.1.1 0.0.0.0 area 0
  Passive Interface(s):
    GigabitEthernet0/0/0
  Routing Information Sources:
    Gateway         Distance      Last Update
    172.16.1.1       110          00:06:41
    172.16.2.1       110          00:06:07
    172.16.3.1       110          00:04:22
  Distance: (default is 110)
```

```
Router-Lisboa# show ip protocols
```

```
Routing Protocol is "ospf 200"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 172.16.2.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.0.0 0.0.255.255 area 0
    172.16.2.1 0.0.0.0 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    172.16.1.1       110          00:07:16
    172.16.2.1       110          00:06:41
    172.16.3.1       110          00:04:57
  Distance: (default is 110)
```

```
Router-Porto# show ip protocols
```

```
Routing Protocol is "ospf 300"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 172.16.3.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.3.0 0.0.0.255 area 0
    192.168.4.0 0.0.0.255 area 0
    172.16.3.1 0.0.0.0 area 0
  Passive Interface(s):
    GigabitEthernet0/0/0
  Routing Information Sources:
    Gateway         Distance      Last Update
    172.16.1.1       110          00:08:21
    172.16.2.1       110          00:07:47
    172.16.3.1       110          00:06:02
  Distance: (default is 110)
```

Other configuration commands to test:

```
Router-XXXXXX# show ip ospf
```

apresenta um resumo da informação associada ao OSPF, nomeadamente os processos OSPF, a identificação do router, as áreas, estatísticas, etc.

```
Router-XXXXXX# show ip ospf database
```

apresenta a base de dados do estado das ligações que formam a topologia de rede.

```
Router-XXXXXX# show ip ospf interface
```

apresenta informação OSPF relativa às interfaces

```
Router-XXXXXX# show ip ospf neighbor
```

apresenta a lista de vizinhos OSPF e o estado das adjacências

You can check the adjacency construction messages in real time using the debug command:

```
Router-XXXXXX# debug ip ospf adj
```

```
Router-XXXXXX# no debug ip ospf adj
```

5 – Verification of neighborhood and adjacent relationships

Example for the Faro Router:

```
Router-Faro#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
172.16.2.1	0	FULL/ -	00:00:39	192.168.2.2	Serial0/1/0

Explanation of the various fields:

Neighbor ID: 172.16.2.1

Identifies the Router ID of the OSPF neighbor, which in this case is the Lisbon router. The Router ID is a unique identifier for each OSPF router on the network, and in this case has been associated with the highest address defined in loopback interfaces.

Pri: 0

This parameter indicates the OSPF priority of the neighbor on the specific interface. Priority value is used in the process of electing the Designated Router (DR) / Backup Designated Router (BDR) in multiple access networks (such as Ethernet). Priority 0 means that this router is not eligible to become a Designated Router (DR) or Backup Designated Router (BDR). In this case, since it is a point-to-point (P2P) serial connection, the DR/BDR election is not relevant, but the priority is displayed.

State: FULL/ -

The FULL state indicates that the OSPF adjacency between this router and its neighbor is fully established. This means that the routers have successfully exchanged and synchronized their OSPF databases (LSDB). In the status column, the symbol “-” means that the adjacency is not associated with a DR or BDR. In point-to-point connections, a DR or BDR is not necessary, so this column is left blank.

Dead Time: 00:00:39

This parameter denotes the time remaining before the OSPF neighbor relationship is considered out of service if no OSPF Hello packets are received from this neighbor within this interval. The Inactivity Interval is typically 4 times the Hello Interval.

Address: 192.168.2.2

This is the IP address of the OSPF neighbor interface that is directly connected to the source router interface (Faro). It is the IP address at the other end of the connection, which in this case is the Lisbon router.

Interface: Serial0/1/0

This is the local interface of the Faro router through which the OSPF neighbor relationship is established. In this case, it is a serial interface Serial0/1/0.

Example for the Lisbon Router:

```
Router-Lisboa#sh ip ospf neighbor
```

Neighbor ID	Pri	State		Dead Time	Address	Interface
172.16.1.1	0	FULL/	-	00:00:31	192.168.2.1	Serial0/1/0
172.16.3.1	0	FULL/	-	00:00:34	192.168.3.2	Serial0/1/1

Since the Lisbon router is centrally located in relation to the Faro and Porto routers, in this case two neighborhood relationships are indicated, respectively with the Faro router (172.16.1.1) and the Porto router (172.16.3.1). The remaining parameters follow the same explanation as above.

Example for the Porto router:

```
Router-Porto#sh ip ospf neighbor
```

Neighbor ID	Pri	State		Dead Time	Address	Interface
172.16.2.1	0	FULL/	-	00:00:34	192.168.3.1	Serial0/1/0

In this case, the only neighborhood relationship is established with the Lisbon router, through the 192.168.3.1 interface.

6 – Verification of connectivity from PCs

Connectivity of the A1 machine with its own network interface:

```
C:\ > ping 192.168.1.101
```

Pinging 192.168.1.101 with 32 bytes of data:

```
Reply from 192.168.1.101: bytes=32 time=1ms TTL=128
Reply from 192.168.1.101: bytes=32 time=18ms TTL=128
Reply from 192.168.1.101: bytes=32 time=20ms TTL=128
Reply from 192.168.1.101: bytes=32 time=20ms TTL=128
```

Ping statistics for 192.168.1.101:

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 20ms, Average = 14ms
```

Connectivity between machine A1 and machine B1:

```
C:\ > ping 192.168.4.101

Pinging 192.168.4.101 with 32 bytes of data:

Reply from 192.168.4.101: bytes=32 time=37ms TTL=125
Reply from 192.168.4.101: bytes=32 time=2ms TTL=125
Reply from 192.168.4.101: bytes=32 time=2ms TTL=125
Reply from 192.168.4.101: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.4.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 37ms, Average = 10ms
```

Connectivity between machine A1 and machine B2:

```
C:\ > ping 192.168.4.102

Pinging 192.168.4.102 with 32 bytes of data:

Reply from 192.168.4.102: bytes=32 time=43ms TTL=125
Reply from 192.168.4.102: bytes=32 time=38ms TTL=125
Reply from 192.168.4.102: bytes=32 time=37ms TTL=125
Reply from 192.168.4.102: bytes=32 time=3ms TTL=125

Ping statistics for 192.168.4.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 43ms, Average = 30ms
```

Connectivity between machine A1 and machine B3:

```
C:\ > ping 192.168.4.103
```

```
Pinging 192.168.4.103 with 32 bytes of data:
```

```
Reply from 192.168.4.103: bytes=32 time=43ms TTL=125
```

```
Reply from 192.168.4.103: bytes=32 time=2ms TTL=125
```

```
Reply from 192.168.4.103: bytes=32 time=2ms TTL=125
```

```
Reply from 192.168.4.103: bytes=32 time=2ms TTL=125
```

```
Ping statistics for 192.168.4.103:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 2ms, Maximum = 43ms, Average = 12ms
```

7 – Manual configuration of the Router ID (RID)

Using the show ip ospf database command, start by checking the RID of the Lisbon router.

```
Router-Lisboa# show ip ospf database
Router-Lisboa#sh ip osp
Router-Lisboa#sh ip ospf d
Router-Lisboa#sh ip ospf database
      OSPF Router with ID (172.16.2.1) (Process ID 200)

      Router Link States (Area 0)

Link ID        ADV Router    Age          Seq#          Checksum Link count
172.16.1.1     172.16.1.1    766         0x80000004   0x00b1b1 4
172.16.2.1     172.16.2.1    731         0x80000005   0x00c3dd 5
172.16.3.1     172.16.3.1    627         0x80000006   0x0077dd 4
```

As can be seen, the RID is given by the value of the loopback interface that was configured earlier, which ensures the stability of the RID without the need for new elections whenever a topological change occurs. However, if necessary, it is possible to modify the parameter manually using the following commands in the ospf router configuration:

```
Router-Lisboa# conf t
Router- Lisboa (config)# router ospf 200
Router- Lisboa (config-router)# router-id 2.2.2.2
Router- Lisboa (config-router)# Reload or use "clear ip ospf process"
command, for this to take effect
Router- Lisboa (config-router)# ^Z
Router-Lisboa# clear ip ospf process
Router-Lisboa#
%SYS-5-CONFIG_I: Configured from console by console
sh ip ospf database
      OSPF Router with ID (2.2.2.2) (Process ID 200)

      Router Link States (Area 0)

Link ID        ADV Router    Age          Seq#          Checksum Link count
172.16.1.1     172.16.1.1    90          0x80000005   0x009e7b 4
172.16.2.1     172.16.2.1    947         0x80000005   0x00c3dd 5
2.2.2.2        2.2.2.2      82          0x80000009   0x006f9d 5
172.16.3.1     172.16.3.1    82          0x80000008   0x0062a8 4
```

8 – OSPF Authentication

OSPF can be configured with authentication to prevent anyone outside the network from sending OSPF messages with illegitimate information. For example, with OSPF messages, an attacker can route all messages to their router and create a denial of service.

If authentication is implemented on a router's interfaces, establishing OSPF neighbor relationships with other routers depends on the consistency of the following parameters:

- **OSPF Hello and Dead Intervals:** The intervals at which OSPF Hello packets are sent and the timeout period after which a neighbor is considered down.
- **OSPF Network Type and Hello Packet Format:** The type of network and the format of the Hello packets (broadcast, point-to-point, etc.).
- **Authentication Method and Key:** The type of authentication used and the specific key or password for that authentication.

As can be seen, if the authentication keys are not identical on both sides, OSPF adjacency will not be established.

OSPF can be configured without authentication or with two types of authentications: simple password and MD5.

- **Plaintext Authentication:**

```
Router(config-if)# ip ospf authentication
Router(config-if)# ip ospf authentication-key [password]
```

- **Security Risk:** Plaintext authentication sends OSPF authentication keys in clear text, which can be intercepted and potentially exploited by attackers. It does not provide strong security.
- **Simple Configuration:** It is straightforward to configure but not recommended for secure environments due to the lack of encryption.

- **MD5 Authentication (Message Digest 5):**

```
Router(config-if)# ip ospf authentication message-digest
Router(config-if)# ip ospf message-digest key [key-id] md5 [password]
```

- **Enhanced Security:** MD5 authentication adds a layer of security by hashing the OSPF packets using MD5, which makes it much harder for an attacker to forge or tamper with routing updates.
- **Key Management:** Requires managing and configuring MD5 keys (key-id and password) on all OSPF routers in the same area to ensure they match. If keys do not match, OSPF neighbors will not form or maintain adjacencies.
- **Complexity:** Slightly more complex to configure compared to plaintext authentication but significantly more secure.

Configuring any of the authentication processes is very simple. Just enable the authentication type and choose a password. As an example, consider configuring serial port 0/1/0 with simple authentication and serial port 0/1/1 with MD5 authentication.

Example application for the Lisbon router, using the two defined methods:

```
Router-Lisboa# conf t

Router-Lisboa (config)# interface serial0/1/0
Router-Lisboa (config-if)# ip ospf authentication
Router-Lisboa (config-if)# ip ospf authentication-key chave1

Router-Lisboa (config)# interface serial0/1/1
Router-Lisboa (config-if)# ip ospf authentication message-digest
Router-Lisboa (config-if)# ip ospf message-digest key 1 md5 chave2
```

If you do not define keys consistently on the remaining routers and their respective interfaces, check that the OSPF adjacencies are not maintained.

```
Router-Lisboa>en
Router-Lisboa#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router-Lisboa(config)#interface serial0/1/0
Router-Lisboa(config-if)#ip ospf authentication
Router-Lisboa(config-if)#ip ospf authentication-key chave1
Router-Lisboa(config-if)#interface serial0/1/1
Router-Lisboa(config-if)#ip ospf authentication message-digest
Router-Lisboa(config-if)#
02:12:36: %OSPF-5-ADJCHG: Process 200, Nbr 172.16.1.1 on Serial0/1/0 from FULL to DOWN,
Neighbor Down: Dead timer expired

02:12:36: %OSPF-5-ADJCHG: Process 200, Nbr 172.16.1.1 on Serial0/1/0 from FULL to DOWN,
Neighbor Down: Interface down or detached
```